File

Name: M39-K01-runin0
Description: KISSsoft example
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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	75.000	
Speed (1/min)	[n]	440.8	145.0
Number of load cycles (in mio.)	[NL]	528.960	174.000
Torque (Nm)	[T]	1624.8	4939.3
Application factor	[KA]	1.25	
Required service life (h)	[H]	20000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	303.000	
Center distance tolerance		ISO 286:2010 Measure js6	
Normal module (mm)	[mn]	6.0000	
Transverse module (mm)	[mt]	6.0000	
Normal diametral pitch (1/in)	[Pnd]	4.23333	
Transverse diametral pitch (1/in)	[Ptd]	4.23333	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	0.0000	
Number of teeth	[z]	25	76
Facewidth (mm)	[b]	44.00	44.00
Hand of gear		Spur gear	
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	77.58	377.50
Shaft diameter (mm)	[ds]	77.58	377.50
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HRC 61	HRC 61
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	Klüberoil GEM 1-220 N with details about wear coefficient kw	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	220.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	19.00
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	80.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	76.491
(mm)	[a0.e/i]	76.360 /76.292
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	2.469 / 2.476

Reference profile Gear 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000

Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	226.509
(mm)	[a0.e/i]	226.330 /226.248
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	2.482 / 2.487

4.1.1 Information on final machining

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	none (only running-in)		
Tip relief by running in (µm)	[Ca L/R]	2.0 / 2.0	2.0 / 2.0

4.2 Basic data

Overall transmission ratio	[itot]	-3.040
Gear ratio	[u]	3.040
Transverse module (mm)	[mt]	6.000
Transverse pressure angle (°)	[αt]	20.000
Working pressure angle (°)	[αwt]	20.000
(°)	[αwt.e/i]	20.008 /19.992
Working pressure angle at normal section (°)	[αwn]	20.000
Base helix angle (°)	[βb]	0.000
Helix angle at operating pitch circle (°)	[βw]	0.000
Reference center distance (mm)	[ad]	303.000
Sum of profile shift coefficients	[Σxi]	0.0000

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient	[x]	0.2485	-0.2485
Generating profile shift coefficient	[xE.e/i]	0.2267/ 0.2153	-0.2783/-0.2920
Virtual gear no. of teeth	[zn]	25.000	76.000
Involute length (mm)	[l_dFa-l_dFf]	11.935	11.698

4.3 Diameters and their allowances

		------- Gear 1	---------- Gear 2 -------
Reference diameter (mm)	[d]	150.000	456.000
Base diameter (mm)	[db]	140.954	428.500
Tip alteration (mm)	[k*mn]	0.000	0.000
Tip diameter (mm)	[da]	164.982	465.018
(mm)	[da.e/i]	164.982 /164.882	465.018 /464.918
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.100	0.000 / -0.100
Tip form diameter (mm)	[dFa]	164.982	465.018
(mm)	[dFa.e/i]	164.982 /164.882	465.018 /464.918
Active tip diameter (mm)	[dNa]	164.982	465.018
(mm)	[dNa.e/i]	164.982 /164.882	465.018 /464.918
V-Circle diameter (mm)	[dv]	152.982	453.018
(mm)	[dv.e/i]	152.721 /152.584	452.661 /452.496

Operating pitch diameter (mm)	[dw]	150.000	456.000
(mm)	[dw.e/i]	150.008 /149.992	456.024 /455.976
Root diameter (mm)	[df]	137.982	438.018
(mm)	[df.e/i]	137.721 /137.584	437.661 /437.496
Active root diameter (mm)	[dNf]	143.447	445.399
(mm)	[dNf.e/i]	143.513 /143.430	445.478 /445.374
Root form diameter (mm)	[dFf]	143.143	442.935
(mm)	[dFf.e/i]	143.012 /142.945	442.672 /442.551

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	1.500	1.500
Effective tip clearance (mm)	[c.e/i]	1.827 / 1.663	1.765 / 1.615
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.284 / 0.209	1.463 / 1.351
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	7.491	4.509
(mm)	[ha.e/i]	7.491 / 7.441	4.509 / 4.459
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	6.009	8.991
(mm)	[hf.e/i]	6.140 / 6.208	9.170 / 9.252
Tooth height (mm)	[h]	13.500	13.500

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	34.851 /34.773	24.153 /24.119
Roll angle to dNf (°)	[ξNf.e/i]	10.966 /10.786	16.288 /16.237
Roll angle at dFf (°)	[ξFf.e/i]	9.826 / 9.663	14.857 /14.793

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.7517	1.3899
Normal tooth thickness at tip circle (mm)	[san]	3.824	4.923
(mm)	[san.e/i]	3.778 / 3.665	4.831 / 4.729
Normal tooth thickness at tip form circle (mm)	[sFan]	3.824	4.923
(mm)	[sFan.e/i]	3.778 / 3.665	4.831 / 4.729
Normal space width at root circle (mm)	[efn]	0.000	4.921
(mm)	[efn.e/i]	0.000 / 0.000	4.967 / 4.988
Pitch on reference circle (mm)	[pt]	18.850	18.850
Base pitch (mm)	[pbt]	17.713	17.713
Transverse pitch on contact-path (mm)	[pet]	17.713	17.713

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	1.056	0.757
Specific sliding at the tip	[ζa]	0.534	0.552
Specific sliding at the root	[ζf]	-1.232	-1.145
Mean specific sliding	[ζm]	0.541	

Sliding factor on tip	[Kga]	0.305	0.219
Sliding factor on root	[Kgf]	-0.219	-0.305

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	44.000
Transverse contact ratio	[εα]	1.669
	[εα.e/m/i]	1.671 / 1.662/ 1.653
Overlap ratio	[εβ]	0.000
Total contact ratio	[εγ]	1.669
	[εγ.e/m/i]	1.671 / 1.662/ 1.653
Length of path of contact (mm)	[gα]	29.555
(mm)	[gα.e/m/i]	29.602 /29.443/29.284
Approach path of contact (mm)	[gf]	12.338
(mm)	[gf.e/m/i]	12.373 /12.273/12.174
Recess path of contact (mm)	[ga]	17.218
(mm)	[ga.e/m/i]	17.229 /17.170/17.110

		Gear 1	Gear 2
Addendum contact ratio	[ε]	0.972	0.697
	[ε.e/i]	0.973 / 0.966	0.699 / 0.687
Length T1-A and T2-A (mm)	[T1A,T2A]	13.314	90.318
(mm)	[.e/i]	13.267 / 13.489	90.318 / 90.190
Length T1-B and T2-B (mm)	[T1B,T2B]	25.156	78.476
(mm)	[.e/i]	25.156 / 25.060	78.429 / 78.619
Length T1-C and T2-C (mm)	[T1C,T2C]	25.652	77.981
(mm)	[.e/i]	25.640 / 25.663	77.945 / 78.016
Length T1-D and T2-D (mm)	[T1D,T2D]	31.027	72.606
(mm)	[.e/i]	30.980 / 31.202	72.606 / 72.477
Length T1-E and T2-E (mm)	[T1E,T2E]	42.869	60.763
(mm)	[.e/i]	42.869 / 42.773	60.716 / 60.906
Length T1-T2 (mm)	[T1T2]	103.632	
(mm)	[.e/i]	103.585 /103.679	
Diameter of single contact point B (mm)	[d-B]	149.664	456.340
(mm)	[d-B.e/i]	149.664 /149.600	456.307 /456.438
Diameter of single contact point D (mm)	[d-D]	154.008	452.436
(mm)	[d-D.e/i]	153.971 /154.150	452.436 /452.353

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	21663.56	
Axial force (N)	[Fa]	0.00	0.00
Radial force (N)	[Fr]	7884.89	7884.89
Normal force (N)	[Fnorm]	23053.88	23053.88
Nominal circumferential force per mm (N/mm)	[w]	492.35	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	21663.56	
Axial force (N)	[Faw]	0.00	0.00
Radial force (N)	[Frw]	7884.89	

Circumferential speed reference circle (m/s)	[v]	3.46	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	3.46	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.8	
Running-in value (µm)	[yf]	0.9	
Correction factor	[CM]	0.800	
Gear blank factor	[CR, bs/b, sr/mn]	0.898	(0.250,5.000)
Basic rack factor	[CB]	0.975	
Material coefficient	[E/Est]	1.000	
Single stiffness (N/mm/µm)	[c']	12.549	
Theoretical single stiffness (N/mm/µm)	[c'th]	17.916	
Contact stiffness (N/mm/µm)	[cγα]	18.841	
Contact stiffness (N/mm/µm)	[cγβ]	16.015	
Reduced mass (kg/mm)	[mRed]	0.06255	
Resonance speed (min-1)	[nE1]	6629	
Resonance ratio (-)	[N]	0.066	
Subcritical range			

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.9	
Bearing distance l of pinion shaft (mm)	[l]	88.000	
Distance s of pinion shaft (mm)	[s]	8.800	
Outside diameter of pinion shaft (mm)	[dsh]	44.000	
Load in accordance with Figure 13, ISO 6336-1:2006	[-]	4	
0:a), 1:b), 2:c), 3:d), 4:e)			
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']	-0.60	
With stiffening			
Effective flank line deviation (µm)	[Fβy]	4.25	
from deformation of shaft (µm)	[fsh·B1]	2.98	
fsh (µm)	[fsh]	2.98	
B1	[B1]	1.00	
fHβ5 (µm)	[fHβ5]	7.50	
Tooth without flank line modification			
Position of contact pattern:	favorable		
from production tolerances (µm)	[fmα*B2]	14.14	
B_2=1.00			
Tooth trace deviation, theoretical (µm)	[Fβx]	5.00	
Running-in value (µm)	[yβ]	0.75	

5.4 K factors

Dynamic factor	[Kv]	1.025
Face load factors		
- Flank	[KHβ]	1.054
- Tooth root	[KFβ]	1.038
- Scuffing	[KBβ]	1.054
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000
Application factor	[KA]	1.250
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.2153	-0.2920
Tooth form factor	[YF]	1.246	1.457
Stress correction factor	[YS]	2.110	1.911
Load application angle (°)	[αFen]	20.411	19.108
Load distribution influence factor	[fε]	1.000	
Load application diameter (mm)	[d_{en}]	154.008	456.340
Bending moment arm (mm)	[hFe]	5.605	6.790
(-)	[hFe*]	0.934	1.132
Tooth root thickness at critical cross section (mm)	[sFn]	12.708	12.989
(-)	[sFn*]	2.118	2.165
Root fillet radius at critical cross section (mm)	[ρF]	2.951	3.329
(-)	[ρF*]	0.492	0.555
Tangent contact point (mm)	[x,y]	6.354 /69.593	
Tangent contact point (mm)	[x,y]		6.494 /219.953
Diameter at critical cross section (mm)	[d_{sFn}]	139.764	440.099
Tangent at critical cross section (°)	[α_{sFn}]	30.000	30.000
Notch parameter	[q_s]	2.153	1.951
Helix angle factor	[Yβ]	1.000	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	44.00	44.00
Nominal stress at tooth root (N/mm²)	[σF0]	215.79	228.47
Tooth root stress (N/mm²)	[σF]	286.93	303.79
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.997	0.995
Surface factor	[YRrelT]	0.957	0.957
Size factor, tooth root	[YX]	0.990	0.990
Life factor	[YNT]	0.902	0.922
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.851	0.868
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	850.00	850.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	516.71	527.25
Limit strength tooth root (N/mm²)	[σFG]	723.40	738.15

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	2.52	2.43
Transmittable power (kW)	[kWRating]	135.06	130.17

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.495	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.882	
Helix angle factor	[Zβ]	1.000	
Effective facewidth (mm)	[beff]	44.00	
Nominal contact stress (N/mm²)	[σH0]	871.80	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1012.87	

Single tooth contact factor	[ZB,ZD]	1.01	1.00
Contact stress (N/mm²)	[σHB, σHD]	1019.55	1012.87
Lubricant coefficient for N_L	[ZL]	1.020	1.020
Speed factor at N_L	[ZV]	0.974	0.974
Roughness factor for N_L	[ZR]	0.980	0.980
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.930	0.962
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.906	0.937
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1358.90	1406.05
Pitting stress limit (N/mm²)	[σHG]	1358.90	1406.05

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.34	1.39
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.33	1.39
Safety regarding transmittable torque	[$SHBD^2$]	1.78	1.93
Transmittable power (kW)	[kWRating]	133.24	144.53

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018
Lubricant load according to FVA Info sheet 54/7 10, Klüberoil GEM 1-220 N with details about wear coefficient kw
Reference data FZG-C Test:

Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	127.666
Contact temperature (°C)	[θBRef,A]	251.735
Lubrication gap thickness (µm)	[hRef,A]	0.051
Specific film thickness in test	[λGFT]	0.102

		------- Gear 1	---------- Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	1.000	
Permissible specific film thickness	[λGFP]	0.143	
Interim results in accordance with ISO/TS 6336-22:2018			
Coefficient of friction	[μ_m]	0.061	
Lubricant factor	[XL]	1.000	
Roughness factor	[XR]	0.924	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tooth mass temperature (°C)	[θM]	84.456	
Tip relief factor	[XCa(A)]	1.000	
Loss factor	[HV]	0.127	
Equivalent Young's modulus (N/mm²)	[Er]	226373.626	
Pressure-viscosity coefficient (m²/N)	[α38]	0.02162	
Dynamic viscosity (Ns/m²)	[ηtM]	25.720	
Roughness average value (µm)	[Ra]	0.600	

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes	0	0
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)	0	

Diameter (mm)	[dy]	143.447	465.018
Relative radius of curvature (mm)	[ρred]	11.603	
Load sharing factor	[XY]	0.333	
Contact stress (N/mm²)	[pH]	736.406	
Contact stress (N/mm²)	[pdyn]	855.569	
Minimal specific film thickness	[λGFY]	0.238	(h_Y=0.143 µm)
Safety against micropitting	[SΛ(B)]	1.667	

9 Safety of hardened layer (subsurface fatigue)

CHD according to	DNV-CG-0036:2021		
		------- Gear 1 ----------	Gear 2 -------
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	0.50 / 0.60
Required hardness at t550 (HV)	[HV.t]	550	550
Required hardening depth CHD (mm)	[t400]	0.90 / 1.10	0.90 / 1.10
Required hardness at t400 (HV)	[HV.t]	400	400
Required hardening depth CHD (mm)	[t300]	1.30 / 1.60	1.30 / 1.60
Required hardness at t300 (HV)	[HV.t]	300	300
Condition for hardness on the surface (HV)	[> HV.0]	460	460
Condition for hardness on the surface (HRC)	[> HRC.0]	46	46

HV.0 is calculated with t_z/a_H = 0.5, S_H = min(S_{EHT}, S_{Hssmin})

Safety of the hardened layer	[SEHT]	1.29	1.29
Required safety	[SHssmin]	1.00	1.00
Result		2	2

-1:Not checked, 0:CHD too shallow, 1:CHD too deep or HV.0 too large, 2:Check performed,
3:Check performed with insufficient hardening depths (t_z/a_H<0.5, t_z/a_{Hst}<0.6)

10 Tooth flank fracture

Calculation method according to ISO/TS 6336-4 (2019)

Calculation according to Method B

		------- Gear 1 ----------	Gear 2 -------
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)		2	2
Diameter (mm)	[dy]	149.664	456.340
Relative radius of curvature (mm)	[ρred]	19.050	19.050
Load sharing factor	[XCP]	1.000	1.000
Contact stress (N/mm²)	[pH]	995.461	995.461
Contact stress (N/mm²)	[pdyn]	1156.543	1156.543
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	
Required hardening depth CHD (mm)	[t550]		0.50 / 0.60
The calculation will be performed with the minimum hardening depth			
Core hardness (HV)	[HV_{core}]	342	342
Hardness curve from:			
HV(y) gear 1		ISO/TS 6336-4, chapter 7.3.3, Method C1	
HV(y) gear 2		ISO/TS 6336-4, chapter 7.3.3, Method C1	
Maximum material exposure	[AFFmax]	0.791	0.791
Depth of maximum load (mm)	[y]	-1.100	-1.100
Material coefficient	[$K_{material}$]	1.131	1.131
Material shearing strength (N/mm²)	[τper]	161.978	161.978
Equivalent stress state (N/mm²)	[τeff]	121.596	121.596
Safety factor according to ISO DTS 19042-1	[SFF]	1.209	1.209

(S_{FF} = 1/($A_{FF,max}$ + 0.2) + 0.2)

11 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	2.00	2.00
Optimal tip relief (µm)	[Ceff]	49.04	
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.000	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	14
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	44.000	
Applicable circumferential force/facewidth (N/mm)	[wBt]	664.588	
$K_{B\gamma}$ = 1.000, $w_{Bt} \cdot K_{B\gamma}$ = 664.588			
Angle factor	[Xαβ]	0.978	
ε_1 =0.972, ε_2 =0.697			

11.1 Flash temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]	0.844		
Tooth mass temperature (°C)	[θMi]	88.92		
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$				
Average flash temperature (Δ°C)	[θflm]	18.98		
Scuffing temperature (°C)	[θS]	451.99		
Γ coordinates (point of highest temperature)	[Γ]	0.625		
[Γ.A]= -0.481, [Γ.E]= 0.671				
Maximum flash temperature (Δ°C)	[θflamax]	29.73		
Maximum contact temperature (°C)	[θBmax]	118.65		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Approach factor	[XJ]	1.000		
Load sharing factor	[XΓ]	0.367		
Dynamic viscosity (mPa*s)	[ηM]	30.07	(80.0	°C)
Coefficient of friction	[μ_m]	0.076		
Required safety	[SBmin]	2.000		
Margin of safety for scuffing, flash temperature	[SB]	9.624		

11.2 Integral temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]	1.000		
Tooth mass temperature (°C)	[θMC]	94.31		
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$				
Integral scuffing temperature (°C)	[θSint]	455.17		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Running-in factor, newly manufactured	[XE]	1.933		
Contact ratio factor	[Xε]	0.230		
Dynamic viscosity (mPa*s)	[ηOil]	30.07	(80.0	°C)
Mean coefficient of friction	[μ_m]	0.060		
Geometry factor	[XBE]	0.298		
Meshing factor	[XQ]	1.000		
Tip relief factor	[XCa]	1.111		
Mean flash temperature (Δ°C)	[θflaint]	20.44		
Integral tooth flank temperature (°C)	[θint]	124.97		

Required safety	[SSmin]	1.800
Safety factor for scuffing (intg.-temp.)	[SSint]	3.642
Safety factor for transmitted moment (int.-T.)	[SSL]	8.343

12 Measurements for tooth thickness

12.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.095 / -0.145	-0.130 / -0.190

12.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	8.000
Base tangent length (no backlash) (mm)	[Wk]	65.115	138.213
Base tangent length with allowance (mm)	[Wk.e/i]	65.026 /64.979	138.090 /138.034
(mm)	[ΔWk.e/i]	-0.089 / -0.136	-0.122 / -0.179
Diameter of measuring circle (mm)	[dMWk.m]	155.220	450.192

12.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	10.820	10.000
Effective diameter of ball/pin (mm)	[DMeff]	11.000	10.000
Radial single-ball measurement, no backlash (mm)	[MrK]	84.452	233.180
Radial single-ball measurement (mm)	[MrK.e/i]	84.353 /84.300	233.002 /232.920
Diameter of measuring circle (mm)	[dMMr.m]	153.042	452.604
Diametral measurement over two balls, no backlash (mm)	[MdK]	168.592	466.361
Diametral measurement over two balls (mm)	[MdK.e/i]	168.394 /168.289	466.005 /465.839
Diametral measurement over pins, no backlash (mm)	[MdR]	168.592	466.361
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	168.394 /168.289	466.005 /465.839
Measurement over 3 pins, no backlash (mm)	[Md3R]	168.281	0.000
Measurement over 3 pins (mm)	[Md3R.e/i]	168.083 /167.979	0.000 / 0.000

12.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
Medium tip diameter (mm)	[da.m]	164.932	464.968
Reference chordal height from da.m (mm)	[hac]	7.650	4.522
Tooth thickness at height hac, chord (mm)	[sc]	10.502	8.339
(mm)	[sc.e/i]	10.409 /10.360	8.210 / 8.150
Tooth thickness on reference circle, arc (mm)	[sn]	10.510	8.339
(mm)	[sn.e/i]	10.415 /10.365	8.209 / 8.149

12.5 Backlash

		---------- Pair ----------
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.012 / -0.012
Radial backlash (mm)	[jrw.e/i]	0.479 / 0.294
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.347 / 0.213
Normal backlash (mm)	[jnw.e/i]	0.326 / 0.200
Center distance allowances (mm)	[Aa.e/i]	0.016 / -0.016
Backlash free center distance (mm)	[aControl.e/i]	302.690 /302.537

Backlash free center distance, allowances (mm)	[jta]	-0.310 / -0.463

		------- Gear 1	---------- Gear 2 -------
dNf.i with aControl (mm)	[dNf0.i]	142.967	444.664
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.023	0.996
Tip clearance (mm)	[c0.i(aControl)]	1.216	1.168

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.0871 / 0.0536	(0°5'14" / 0°3'13")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.2648 / 0.1630	(0°15'53" / 0°9'47")

13 Gear tolerances

		------- Gear 1	---------- Gear 2 -------
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A6
Single pitch tolerance (µm)	[fpT]	±11.00	±11.00
Base pitch tolerance (µm)	[fpbT]	±10.34	±10.34
Sector pitch tolerance (µm)	[Fpz/8T]	±21.00	±27.00
Cumulative pitch tolerance, total (µm)	[FpT]	33.00	41.00
Adjacent pitch difference tolerance (µm)	[fuT]	15.00	16.00
Profile form tolerance (µm)	[ffαT]	12.00	12.00
Profile slope tolerance (µm)	[fHαT]	±9.50	±9.50
Profile tolerance, total (µm)	[FαT]	15.00	15.00
Helix form tolerance (µm)	[ffβT]	11.00	12.00
Helix slope tolerance (µm)	[fHβT]	±10.00	±10.00
Helix tolerance, total (µm)	[FβT]	15.00	16.00
Runout tolerance (µm)	[FrT]	30.00	37.00
Single flank composite tolerance, total (µm)	[FisT]	43.00	51.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	10.00	10.00
Radial composite tolerance, total (µm)	[Fi"]	51.00	60.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	22.00	22.00

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R41	R41
Radial composite tolerance, total (µm)	[FidT]	45.00	60.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	21.00	25.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

Total helix deviation for calculation (µm)	[Fβ]	16.00
Distance between bearings for calculation (mm)	[LG]	88.00
Maximum value for axis deviation error (µm)	[fΣβ]	± 16.00
Maximum value for axis inclination error (µm)	[fΣδ]	± 32.00

14 Modifications and determination of the tooth form

14.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

15 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	307.252

15.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	4.580	16.605
Total mass (kg)	[Σm]	21.185	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.01658	0.71892
System (da+df)/2...di (kg*m²)	[J]	0.09438	

15.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	3.500
Torsion when subjected to nominal torque (°)	[δcr]	0.027
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	78.219

15.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.060
Wear sliding coefficient by Niemann	[ζw]	0.903
Loss factor	[HV]	0.127
Gear power loss (kW)	[PVZ]	0.570
Meshing efficiency (%)	[ηz]	99.241

16 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	20000.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	>	1000000

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	1e+06	1e+06

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

16.1 Damage

Damage relative to the required service life (H, 20000.0 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	0.0000	0.0000	0.0000

17 Reliability calculation

17.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

17.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	β	t_0	T	R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	26448	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	26448	1.300e+00	9.014e+29	4.295e+30	100.0000

2	Tooth root	8700	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	8700	1.300e+00	9.014e+29	4.295e+30	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

17.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	20000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	100.0000
Reliability R, gears subsystem (%)	[R_{subG}]	100.0000
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	> 1'000'000
Service life H, gears subsystem (h)	[H_{subG}]	> 1'000'000

18 Remarks

18.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

18.2 Calculations and factors

- Details of calculation method:
  c_v according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 723)
